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Multichannel dispenser assembly

Abstract

A dispenser assembly includes a housing including a plurality of inlets and an outlet. The housing defines a first portion of a flow path. A rotatable disk is positioned within the housing, and the rotatable disk includes a groove in a top side of the rotatable disk. The groove defines a second portion of the flow path. A positioning wheel is coupled to the rotatable disk. A motor is mechanically coupled to the positioning wheel for driving rotation of the rotatable disk about an axis to align the groove of the rotatable disk with one of the plurality of inlets. When the groove of the rotatable disk engages with one of the plurality of inlets, the first portion of the flow path and the second portion of the flow path mesh to provide the flow path between one of the plurality of inlets and the outlet.

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Background/Summary**FIELD OF THE INVENTION**

(1) The present invention relates generally to washing appliances, such as dishwashing appliances, and more specifically to a multichannel dispenser assembly for such appliances.

BACKGROUND OF THE INVENTION

(2) Domestic appliances may include dishwashers, washing machines, dryers, and other appliances. Certain domestic appliances, hereinafter collectively referred to as “washing appliances,” are configured for washing items such as dishes or clothing articles. Washing appliances, such as washing machine appliances and dishwasher appliances generally include a tub that defines a wash chamber and a pump for directing fluids from a fluid reservoir into the wash chamber during washing operations.

(3) For example, washing appliances generally include a dispenser for dispensing one or more

additives, such as dish detergent, laundry detergent, or other treatment chemicals such as rinse aid, fabric softener, etc., into the wash chamber. Traditionally, such dispensers either mix the fluids or maintain distinct channels. Mixing the fluids may be undesirable in some cases, such as a washing cycle, e.g., a dishwasher cycle, where specific liquids are desired to be dispensed at specific times. Further, maintaining distinct channels may be undesirable because separate control devices may be required for all channels, thereby adding complexity.

(4) Accordingly, a dispenser assembly for selectively dispensing separate wash fluids without separate control devices to maintain distinct channels would be advantageous.

BRIEF DESCRIPTION OF THE INVENTION

(5) Aspects and advantages of the invention will be set forth in part in the following description, or may be obvious from the description, or may be learned through practice of the invention.

(6) In one example embodiment, a washing appliance includes a wash chamber for receipt of articles for washing, a door for selectively opening and closing the wash chamber, and a dispenser assembly. The dispenser assembly includes a housing including a plurality of inlets and an outlet. The housing defines a first portion of a flow path. A rotatable disk is positioned within the housing, and the rotatable disk includes a groove in a top side of the rotatable disk. The groove is configured to selectively align with one of the plurality of inlets. The groove defines a second portion of the flow path. A positioning wheel is coupled to the rotatable disk. The positioning wheel is configured to provide positional feedback of the rotatable disk. A motor is mechanically coupled to the positioning wheel for driving rotation of the rotatable disk about an axis to align the groove of the rotatable disk with one of the plurality of inlets. When the groove of the rotatable disk engages with one of the plurality of inlets, the first portion of the flow path and the second portion of the flow path mesh to provide the flow path between one of the plurality of inlets and the outlet.

(7) In another example embodiment, a dispenser assembly includes a housing including a plurality of inlets and an outlet. The housing defines a first portion of a flow path. A rotatable disk is positioned within the housing, and the rotatable disk includes a groove in a top side of the rotatable disk. The groove is configured to selectively align with one of the plurality of inlets. The groove defines a second portion of the flow path. A positioning wheel is coupled to the rotatable disk. The positioning wheel is configured to provide positional feedback of the rotatable disk. A motor is mechanically coupled to the positioning wheel for driving rotation of the rotatable disk about an axis to align the groove of the rotatable disk with one of the plurality of inlets. When the groove of the rotatable disk engages with one of the plurality of inlets, the first portion of the flow path and the second portion of the flow path mesh to provide the flow path between one of the plurality of inlets and the outlet.

(8) These and other features, aspects and advantages of the present invention will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures.

(2) FIG. 1 provides a front view of a dishwashing appliance in accordance with an embodiment of the present disclosure.

(3) FIG. 2 provides a perspective view of the dishwashing appliance of FIG. 1 with a door of the

dishwashing appliance hidden to expose other components in accordance with aspects of the present disclosure.

(4) FIG. 3 provides a perspective view of a multichannel dispenser assembly in accordance with an embodiment of the present disclosure.

(5) FIG. 4 provides a sectional view of the multichannel dispenser assembly of FIG. 3 along line 4-4.

(6) FIG. 5 provides a top, section view of the multichannel dispenser assembly of FIG. 3.

(7) FIG. 6 provides alternative embodiments of a seal for a multichannel dispenser assembly in accordance with an embodiment of the present disclosure.

(8) FIG. 7 provides a schematic view of an example fluid reservoir of a washing appliance such as the example dishwashing appliance of FIG. 1 in accordance with an embodiment of the present disclosure.

(9) Repeat use of reference characters in the present specification and drawings is intended to represent the same or analogous features or elements of the present invention.

DETAILED DESCRIPTION

(10) Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

(11) As used herein, the term “or” is generally intended to be inclusive (i.e., “A or B” is intended to mean “A or B or both”). The term “article” may refer to, but need not be limited to, dishes, pots, pans, silverware, and other cooking utensils and items that can be cleaned in a dishwashing appliance; garments, fabrics, textiles, linens, clothing and other clothing articles and items that can be cleaned in a laundry appliance. In The term “fluid” refers to a liquid used for washing and/or rinsing the articles and is typically made up of water that may include additives such as detergent or other treatments, such as liquid detergent, powder detergent, bleach, and softener. The terms “first” and “second” may be used interchangeably to distinguish one component from another and are not intended to signify the location or importance of the individual components.

(12) Benefits, other advantages, and solutions to problems are described below with regard to specific embodiments. However, the benefits, advantages, solutions to problems, and any feature(s) that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as a critical, required, or essential feature of any or all the claims.

(13) While described in the context of a dishwashing appliance as an example of a washing appliance, the present disclosure may be used in any suitable washing appliance. For example, aspects of the present disclosure may include different embodiments of a washing appliance including a tub, a wash chamber, and fluids flowing into the wash chamber. Washing appliances may generally include dishwashers, washing machines, and other appliance, e.g., washing machines and dishwashers generally include a tub that defines a wash chamber and a pump for directing fluids from a fluid reservoir into the wash chamber during washing operations.

(14) Referring now to the drawings, FIGS. 1 and 2 illustrate a front view of a washing appliance that may be configured in accordance with aspects of the present disclosure. For the particular example embodiment of FIGS. 1 and 2, the washing appliance is a dishwashing appliance **100**. As stated above, dishwashing appliance **100** is provided for example purposes only and the present subject matter may be used in any suitable washing appliances, such as washing machines and dishwashers. As shown particularly in FIG. 2, dishwashing appliance **100** includes a cabinet **102** having a tub **104** therein that defines a wash chamber **106** for the receipt of articles for washing.

Tub **104** may include a front opening (not shown) and a door **110** for selectively opening and closing the wash chamber **106**. As shown, door **110** may be hinged to wash chamber **106** for movement between a closed position (shown in FIGS. **1** and **2**) and an open position for loading and unloading articles into and from the dishwasher appliance **100**. In addition, as shown, a latch **116** may be used to lock and unlock the door **110** for access to the wash chamber **106**.

(15) As shown in FIG. **2**, the dishwashing appliance **100** may be further equipped with a controller **156** to regulate operation of the dishwashing appliance **100**. Controller **156** may include one or more memory devices and one or more microprocessors, such as general or special purpose microprocessors operable to execute programming instructions or micro-control code associated with a cleaning cycle. The memory may represent random access memory such as DRAM or read only memory such as ROM or FLASH. In an embodiment, the processor executes programming instructions stored in the memory. The memory may be a separate component from the processor or may be included onboard within the processor. Further, controller **156** may be positioned in a variety of locations throughout dishwashing appliance **100**. In the illustrated embodiment, for example, controller **156** may be located within a control panel area **158** of door **110** as shown in FIG. **1**. Typically, controller **156** includes a user interface panel/control **160** through which a user may select various operational features and modes and monitor progress of the dishwashing appliance **100**.

(16) Referring particularly to FIG. **2**, the dishwashing appliance **100** may further include a fluid reservoir **170** containing a plurality of fluids **178** for washing and/or rinsing the articles within the wash chamber **106**. Furthermore, as shown, dishwashing appliance **100** may also include a pump **174** for directing or pumping the plurality of fluids **178** from the fluid reservoir **170** and into the wash chamber **106**. Further, and as will be discussed in more detail herein, the dishwashing appliance **100** may include a dispenser assembly **200** for controlling the plurality of fluids **178** flowing from the fluid reservoir **170** and into the wash chamber **106** along a flow path **172**.

(17) Thus, as shown in FIG. **2**, the dispenser assembly **200** may be in fluid communication with a dispenser **176** in fluid communication with the dispenser assembly **200** for dispensing one of the plurality of different fluids into the wash chamber **106** at a time. As shown in FIG. **2**, the fluid reservoir **170**, the pump **174**, the dispenser **176**, and the dispenser assembly **200** may be disposed within door **110**. As such, FIG. **2** depicts the fluid reservoir **170**, the dispenser assembly **200**, the pump **174**, and the dispenser **176** as having a particular arrangement along the flow path **172**. However, it should be understood that such an arrangement is not limiting, and a variety of arrangements can be implemented in accordance with the present disclosure. For example, the pump **174** and the dispenser **176** can be arranged in a different order along the flow path **172** such that the dispenser **176** is disposed adjacent or connected to the dispenser assembly **200**.

(18) Referring now to FIG. **3**, a perspective view of the dispenser assembly **200** according to an example embodiment of the present disclosure is illustrated. The dispenser assembly **200** can be used with a variety of appliances, e.g., any suitable washing appliance, such as the dishwashing appliance **100** of FIGS. **1** and **2** or another washing appliance. As shown, the dispenser assembly **200** includes a housing **202** which may include a plurality of inlets **208** and an outlet **210**. In particular, the plurality of inlets **208** and outlet **210** may extend from housing **202**, such that the plurality of inlets **208** and outlet **210** extend in a parallel direction. Further, the plurality of inlets **208** may include any suitable number of inlets, such as five (5) separate inlets as shown. In further embodiments, housing **202** of dispenser assembly **200** may have more or less than five (5) separate inlets. In some embodiments, housing **202** of the dispenser assembly **200** may also have a first housing portion **204** detachably secured to a second housing portion **206**. Referring briefly to FIG. **4**, dispenser assembly **200** may also include at least one housing seal **232**. Generally, housing seal **232** may be arranged between the first housing portion **204** and the second housing portion **206**. In particular, housing seal **232** may be positioned within a recess **234** of second housing portion **206**. Further, in some embodiments, the first housing portion **204** and the second housing portion **206**

may be secured together via one or more fasteners **211**. In some embodiments, the fastener(s) **211** can include a variety of different fasteners, such as screws, nuts and bolts, or any other suitable fasteners.

(19) Referring now to FIG. 4, a cross-sectional view of the dispenser assembly **200** of FIG. 3 along line 4-4 is illustrated. As shown, the dispenser assembly **200** includes a rotatable disk **220** positioned within housing **202**. The rotatable disk **220** may have a groove **222** on a top side **221** of the rotatable disk **220**, and the groove **222** may be configured to selectively align with one of the plurality of inlets **208** of the housing **202** to provide a flow path **212** between one of the plurality of inlets **208** and the outlet **210**. For example, each inlet of the plurality of inlets **208** of the housing **202** may define a first portion **214** of flow path **212** and groove **222** of rotatable disk **220** may define a second portion **216** of flow path **212**, such that when the groove **222** of the rotatable disk **220** engages with one of the plurality of inlets **208**, the first portion **214** of the flow path **212** and the second portion **216** of the flow path **212** mesh to provide the flow path **212** between one of the plurality of inlets **208** and the outlet **210**.

(20) Referring to FIGS. 4 and 5, the rotatable disk **220** may have a protrusion **223** extending through the second housing portion **206** and into a positioning wheel **224**. Protrusion **223** may be mechanically coupled to a motor **226**, such as a geared motor, for driving rotation of the positioning wheel **224** and rotatable disk **220** about an axis A. As such, the motor **226** shown in FIG. 4 drives the rotatable disk **220** about the axis A to align the groove **222** of the rotatable disk **220** with one of the plurality of inlets **208**. In general, positioning wheel **224** may be configured to provide positional feedback of the rotatable disk **220**. As shown in FIG. 5, positioning wheel **224** may include a ramp **227**, or a plurality of ramps **227** circumferentially spaced around positioning wheel **224**. For example, each ramp **227** may be configured to engage with a switch **225**, in electrical communication with controller **156**, to provide positional feedback of the rotatable disk **220**. In particular, each ramp **227** may be circumferentially differing lengths, such as to differentiate positions of positioning wheel **224** and thus rotatable disk **220**.

(21) Referring to FIG. 6, dispenser assembly **200** may also include a groove seal **228** disposed around groove **222**. The groove seal **228** may be configured for engaging housing **202**, such as first housing portion **204**, and fluidly sealing flow path **212**, e.g., groove seal **228** may fluidly seal between first housing portion **204** and rotatable disk **220**. In particular, groove seal **228** may be positioned in a groove seal recess **229** around groove **222**. Furthermore, a plurality of inlet seals **230** may be disposed within rotatable disk **220** for fluidly separating each of the plurality of inlets **208** of first housing portion **204**. For example, the plurality of inlet seals **230** may be configured to engage with each of the plurality of inlets **208** when inlets **208** are not engaging with groove **222**. In particular, each inlet seal of the plurality of inlet seals **230** may be positioned within a respective recess of a plurality of recesses **236** formed into the rotatable disk **220**. The plurality of recesses **236** may be positioned in the rotatable disk **220** relative to the position of the plurality of inlets **208** of the housing **202**, when the dispenser assembly **200** is assembled. In certain embodiments, such as shown in FIG. 6, the inlet seal(s) **230** and/or the housing seal **232** may be O-rings or any other suitable flexible seal.

(22) Referring now to FIG. 7, in example embodiments, the dispenser assembly **200** may be configured to fluidly couple with a fluid reservoir **170** containing a plurality of different fluids **178**. For example, each fluid of the plurality of different fluids **178** may be stored within one of a plurality of separate compartments **242** defined within fluid reservoir **170**. In such embodiments, the dispenser assembly **200** is fluidly coupled with the fluid reservoir **170** containing the plurality of fluids **178** via a plurality of conduits **240** coupled to the plurality of inlets **208** of the housing **202**. For example, each conduit of the plurality of conduits **240** may be in fluid communication with a respective one of the separate compartments **242**. In other words, each fluid of the plurality of fluids **178** is different from every other fluid, and each respective one different fluid is in communication with a respective one conduit of the plurality of conduits **240**, such that each

conduit is in communication with one and only one fluid. Thus, for example, one of the plurality of fluids **178** may flow into the dispenser assembly **200** through one conduit of the plurality of conduits **240** and into one inlet of the plurality of inlets **208**. Further, the outlet **210** of housing **202** is configured for being connected to a pump of a washing appliance, such as the pump **174** of the dishwashing appliance **100** shown in FIG. 1, to be provided to a dispenser of the washing appliance, such as the dispenser **176** of the dishwashing appliance **100**. As shown in FIG. 7, a pump line **244** may be connected between outlet **210** of the housing **202** and the pump, e.g., pump **174** of the dishwashing appliance **100**, for the different fluids **178** to be pumped therethrough. As such, dispenser **176** may be configured for dispensing one of the plurality of different fluids **178** into the appliance at a time. In some example embodiments, one of the plurality of fluids **178** may be one of water or air, whereby dispenser assembly may be configured to run backwards to clean out groove **222** of any residual fluids.

(23) As may be seen from the above, a multi-port disk diverter may be positioned in a dishwasher appliance. The diverter may be driven by a motor and have a positional feedback mechanism. The diverter disk may include a fluid flow channel which is associated with one outlet and multiple inlets present in the stationary housing. The disk may rotate about an axis with the help of the motor to align the channel for the fluid flow through one of the inlets to the outlet present in the center of the housing. Below the diverter disk is a second housing portion which may constrain the parts of the disk diverter system. The feedback wheel, feedback switch and motor may control the rotational position of the diverter disk. In addition, seals are located on the disk to prevent leakage while the fluid flows through the channel.

(24) This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A washing appliance, comprising: a tub defining a wash chamber for receipt of articles for washing; a door for selectively opening and closing the wash chamber; and a dispenser assembly, the dispenser assembly comprising: a housing comprising a plurality of inlets and an outlet, the housing defining a first portion of a flow path; a rotatable disk positioned within the housing, the rotatable disk comprising a groove in a top side of the rotatable disk, the groove configured to selectively align with one of the plurality of inlets, the groove defining a second portion of the flow path; a positioning wheel coupled to the rotatable disk, the positioning wheel configured to provide positional feedback of the rotatable disk; and a motor mechanically coupled to the positioning wheel for driving rotation of the rotatable disk about an axis to align the groove of the rotatable disk with one of the plurality of inlets, wherein, when the groove of the rotatable disk engages with one of the plurality of inlets, the first portion of the flow path and the second portion of the flow path mesh to provide the flow path between one of the plurality of inlets and the outlet.
2. The washing appliance of claim 1, further comprising a groove seal disposed around the groove, the groove seal configured for engaging the housing and fluidly sealing the flow path.
3. The washing appliance of claim 2, wherein the rotatable disk comprises a plurality of inlet seals within the rotatable disk, the plurality of inlet seals configured to engage with each of the plurality of inlets when the inlets are not engaging with the groove.
4. The washing appliance of claim 3, wherein the rotatable disk comprises a protrusion extending

through the housing and engaging with one or both of the positioning wheel and the motor.

5. The washing appliance of claim 4, wherein the housing comprises a first housing portion detachably secured to a second housing portion.

6. The washing appliance of claim 5, further comprising a housing seal positioned within a recess formed into the second housing portion.

7. The washing appliance of claim 5, wherein the groove seal is positioned within a recess formed into the rotatable disk around the groove.

8. The washing appliance of claim 1, wherein each inlet seal of the plurality of inlet seals is positioned within a respective recess of a plurality of recesses formed into the rotatable disk, the plurality of recesses positioned in the rotatable disk relative to the position of the plurality of inlets of the housing.

9. The washing appliance of claim 1, further comprising a fluid reservoir containing a plurality of different fluids and a plurality of conduits coupled to the fluid reservoir, whereby each conduit of the plurality of conduits is in fluid communication with a respective one of the plurality of different fluids, wherein the dispenser assembly is configured to fluidly couple with the fluid reservoir via the plurality of conduits coupled to the plurality of inlets.

10. The washing appliance of claim 1, further comprising a fluid reservoir, the fluid reservoir comprising a plurality of separate compartments defined within the fluid reservoir, wherein each separate compartment of the plurality of separate compartments contains a respective one different fluid of a plurality of different fluids, further comprising a plurality of conduits coupled to the fluid reservoir, whereby each conduit of the plurality of conduits is in fluid communication with a respective one of the separate compartments, wherein the dispenser assembly is fluidly coupled to the fluid reservoir at the plurality of inlets of the dispenser assembly via the plurality of conduits, whereby each conduit provides a respective one different fluid of the plurality of different fluids to the dispenser assembly.

11. A dispenser assembly for a washing appliance, the dispenser assembly comprising: a housing comprising a plurality of inlets and an outlet, the housing defining a first portion of a flow path; a rotatable disk positioned within the housing, the rotatable disk comprising a groove in a top side of the rotatable disk, the groove configured to selectively align with one of the plurality of inlets, the groove defining a second portion of the flow path; a positioning wheel coupled to the rotatable disk, the positioning wheel configured to provide positional feedback of the rotatable disk; and a motor mechanically coupled to the positioning wheel for driving rotation of the rotatable disk about an axis to align the groove of the rotatable disk with one of the plurality of inlets, wherein, when the groove of the rotatable disk engages with one of the plurality of inlets, the first portion of the flow path and the second portion of the flow path mesh to provide the flow path between one of the plurality of inlets and the outlet.

12. The dispenser assembly of claim 11, further comprising a groove seal disposed around the groove, the groove seal configured for engaging the housing and fluidly sealing the flow path.

13. The dispenser assembly of claim 12, wherein the rotatable disk comprises a plurality of inlet seals within the rotatable disk, the plurality of inlet seals configured to engage with each of the plurality of inlets when the inlets are not engaging with the groove.

14. The dispenser assembly of claim 13, wherein the rotatable disk comprises a protrusion extending through the housing and engaging with one or both of the positioning wheel and the motor.

15. The dispenser assembly of claim 14, wherein the housing comprises a first housing portion detachably secured to a second housing portion.

16. The dispenser assembly of claim 15, further comprising a housing seal positioned within a recess formed into the second housing portion.

17. The dispenser assembly of claim 15, wherein the groove seal is positioned within a recess formed into the rotatable disk around the groove.

18. The dispenser assembly of claim 11, wherein each inlet seal of the plurality of inlet seals is positioned within a respective recess of a plurality of recesses formed into the rotatable disk, the plurality of recesses positioned in the rotatable disk relative to the position of the plurality of inlets of the housing.

19. The dispenser assembly of claim 11, further comprising a fluid reservoir containing a plurality of different fluids and a plurality of conduits coupled to the fluid reservoir, whereby each conduit of the plurality of conduits is in fluid communication with a respective one of the plurality of different fluids, wherein the dispenser assembly is configured to fluidly couple with the fluid reservoir via the plurality of conduits coupled to the plurality of inlets.

20. The dispenser assembly of claim 11, further comprising a fluid reservoir, the fluid reservoir comprising a plurality of separate compartments defined within the fluid reservoir, wherein each separate compartment of the plurality of separate compartments contains a respective one different fluid of a plurality of different fluids, further comprising a plurality of conduits coupled to the fluid reservoir, whereby each conduit of the plurality of conduits is in fluid communication with a respective one of the separate compartments, wherein the dispenser assembly is fluidly coupled to the fluid reservoir at the plurality of inlets of the dispenser assembly via the plurality of conduits, whereby each conduit provides a respective one different fluid of the plurality of different fluids to the dispenser assembly.
